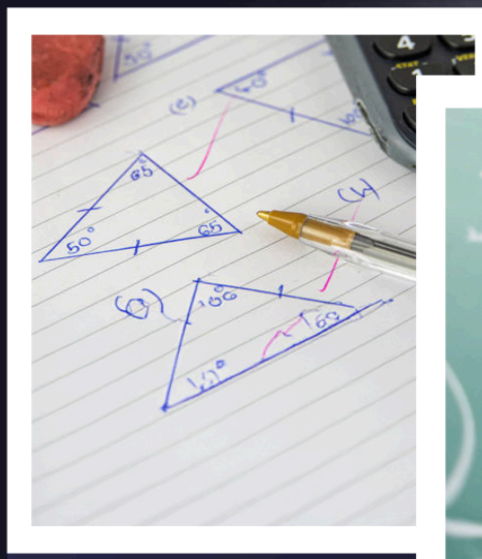
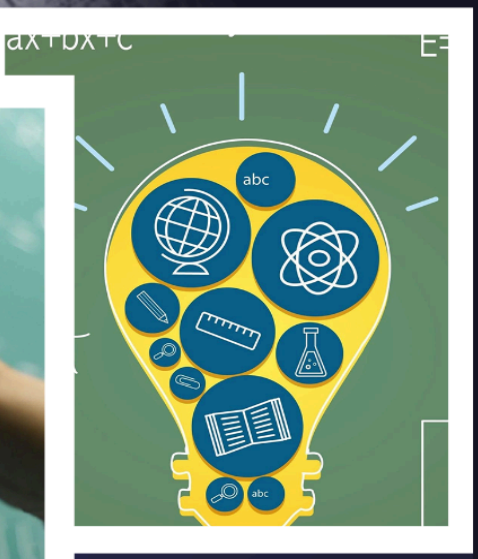
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Mensuration 2D





Mensuration 2D

Triangle

- $Perimeter = a + b + c$
- **Area of Triangle**
  - ➔ Using base and height :

    $$Area = \frac{1}{2} \times Base \times Height = \frac{1}{2} \times b \times h$$

  - ➔ Using Heron's formula :

    $$Area = \sqrt{s(s-a)(s-b)(s-c)}$$

    Here, $s = \frac{a+b+c}{2}$

Equilateral Triangle

- Perimeter = 3a
- Height of equilateral triangle = $\frac{\sqrt{3}}{2}$ a
- Area of equilateral triangle = $\frac{\sqrt{3}}{4} a^2$
- Radius of the incircle (r) of an equilateral triangle = (side × √3) / 6
- The circumradius of an equilateral triangle is = $\frac{a}{\sqrt{3}}$

Isosceles Triangle

- Perimeter = 2a + b
- Area of triangle ABC = $\frac{b}{4}\sqrt{4a^2 - b^2}$ = $\frac{1}{2}a^2 \sin sin \propto$

Scalene Triangle

- Perimeter = a + b + c
- Semi perimeter (s) = $\frac{a+b+c}{2}$
- Area = $\sqrt{\{s(s-a)(s-b)(s-c)\}}$

Square

- $Perimeter = 4 \times a = 4a$
- $Area = a^2$
- $Diagonal\ of\ Square = a\sqrt{2}$

Rectangle

- $Perimeter = 2 \times (Length + Breadth) = 2(l + b)$
- $Area = Length \times Breadth = l \times b$
- $Diagonal\ of\ Rectangle = \sqrt{l^2 + b^2}$

Rhombus

- $Perimeter = 4 \times a = 4a$
- $Area = \frac{1}{2} \times d_1 \times d_2$
- $Diagonals\ of\ Rhombus:$

  ➔ $d_1 = 2 \times \sqrt{a^2 - (\frac{d_2}{2})^2}$

  ➔ $d_2 = 2 \times \sqrt{a^2 - (\frac{d_1}{2})^2}$

Parallelogram

- $Perimeter = 2 \times (a + b)$
- $Area = Base \times Height = b \times h$
- $Area\ of\ a\ parallelogram = a \times b \times sin\theta$

  Where, θ = angle between a and b

- $Diagonal\ of\ Parallelogram = d = \sqrt{a^2 + b^2 - 2abcos\theta}$

Trapezium

- $Area\ of\ a\ Trapezium\ = \frac{1}{2} \times (a + b) \times h$
- $Height\ of\ Trapezium\ = \sqrt{c^2 - (\frac{(b-a)^2 + c^2 - d^2}{2(b-a)})^2}$

Circle

- $Circumference\ of\ circle\ = 2\pi r$
- $Area\ of\ circle\ = \pi r^2$

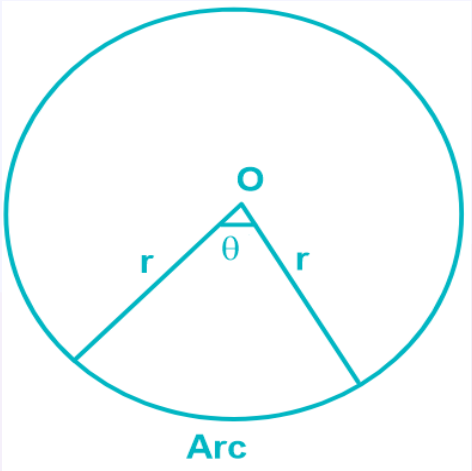
Semi-Circle:

- Diameter = 2r
- Perimeter = r (2 + π)
- Area = $\frac{\pi r^2}{2}$

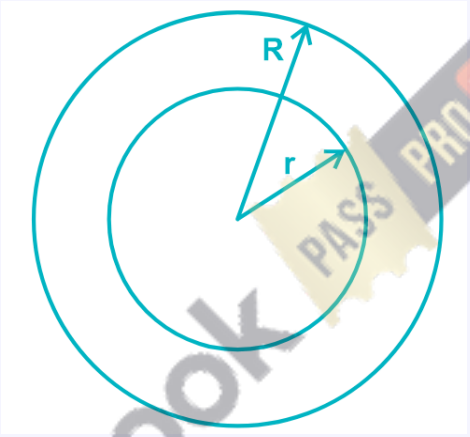
Direct Common Tangent:

- $Length\ of\ Direct\ Common\ Tangent\} = \sqrt{\{d^2 - (r_1 - r_2)^2\}}$
- $Length\ of\ Transverse\ Common\ Tangent = \sqrt{d^2 - (r_1 + r_2)^2}$

Sector:



- Perimeter of shaded part = 2 (r + θ)
- Area of shaded part = $\pi r^2 \left(\frac{\theta}{360}\right)$
- Length of arc = $2 \pi r \left(\frac{\theta}{360}\right)$

Concentric circle:

- Area enclosed by Concentric circle = π (R + r) (R - r)
- Width of Path = R - r